

2017-Ph-H1-Q15

Section: Particles and Waves

Topic: Radiation

A graph shows the number of radioactive nuclei over time. Determine the **half-life**.

Key Concept:

- Half-life = time for number of nuclei to halve
- Find when count goes from e.g. 800 to 400

From the graph, if this occurs over **2.0 minutes**, that's the half-life.

 **Final Answer: C — 2.0 minutes**

Revision Tips:

- Count drops by **half** every half-life
- Graph is always **exponential decay**